Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing with Business Management**

Assessment	**Databases 2.0**
Assessment Type	**Examination**
Date	**Wednesday 28th May 2025**
Time	**09.00am – 12.00pm**

Examination paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

A smart home system allows homeowners to manage various smart devices within their home. The system tracks the people, smart devices installed, the rooms where these devices are located and the interactions between users and devices.

a) Identify **FOUR (4)** entities in the smart home database system based on the scenario provided. **4**

b) The smart home database ensures that when a device is assigned to a room, the room must exist in the system. **4**

Explain in detail how this is enforced in the database and **describe** what could happen if this rule is not applied.

c) Identify **TWO (2)** benefits of using primary keys in the smart home database. **2**

Total 10 Marks

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Marks

Question 2

Revisit the smart home database.

The smart home system enables homeowners to manage various smart devices, such as lights, thermostats and security cameras. The system records device interactions, such as turning devices on or off.

Transactions must be handled correctly to maintain data consistency and avoid system errors.

a) Identify each property in **ACID**. For each ACID property, describe a transaction in the smart home database and explain how the property would apply. **8**

b) Explain the purpose of the **ROLLBACK** command in SQL. Identify a situation where it would be used. **2**

Total 10 marks

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Marks

Question 3

10

In Question 1 and 2, you worked with a smart home system.

Now, consider a different but related system: a smart home energy monitoring system that tracks household energy usage.

Each home contains multiple appliances and each appliance belongs to only **ONE (1)** home.

Energy meters are used to track electricity usage. A single meter can monitor multiple appliances, but each appliance is connected to only **ONE (1)** meter.

Draw an Entity Relationship Diagram for the smart home energy monitoring database.

You need to identify relationships, cardinalities and label all the primary **and** foreign keys. Clearly state which entity the keys belong to.

Total 10 marks

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Marks

Question 4

a) The smart home security system allows homeowners to manage security devices and track security-related events. The system collects various details when an authorised user is added and when security devices log events. **8**

Using the information provided below, identify **EIGHT (8)** attributes that would be suitable for managing smart home security data in a database system.

Smart Home Security System Form

Date: 15 April 2025

User ID: U402

User Name: Jane Doe

Access Level: Admin

Security Devices Linked:

- **Front Door Camera**
- **Motion Sensor (Living Room)**

Security Events Recorded:

- **Front Door Camera** - **Detected Motion** (Timestamp: 10 July 2024, 3:15 AM, Severity: Medium)
- **Motion Sensor (Living Room)** - **Triggered Alarm** (Timestamp: 12 July 2024, 2:30 AM, Severity: High)

Emergency Contact Setup: Yes (Contact: +44 7890 123456)

Notifications Enabled: Yes

Device Location: Home Entrance, Living Room

Event Resolution Status: Reviewed

b) The smart home security database needs to store event logs efficiently. Explain why indexing would be beneficial for querying security events and give **ONE (1)** example of an attribute that should be indexed. **2**

Total 10 marks

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Marks

Question 5

a) The smart home security system manages *users, security devices* and *security events*. The system performs various transactions related to these entities. **6**

Complete the CRUD matrix below to show the following transactions:

Transaction 1 – Register a new user in the security system.

Transaction 2 – Add a new security device to the system.

Transaction 3 – A user updates their personal details.

Transaction 4 – Update a security device's settings, such as its sensitivity level or recording duration.

Transaction 5 – Remove a security device that is no longer in use.

Transaction 6 – Retrieve a history of security events recorded by a specific device.

Transaction	User	Security Device	Security Event
1			
2			
3			
4			
5			
6			

b) In the smart home security database, multiple users can control multiple security devices. **4**

Describe how this relationship should be represented in a relational database. Explain why this structure is necessary.

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Marks

Question 6

Consider the smart home security system.

Write the appropriate SQL command to perform each of the following tasks to the security device data provided below.

security_device

device_id	device_type	location	status (TRUE=Active, FALSE=Inactive)
201	camera	entrance	TRUE
202	motion sensor	living room	TRUE
203	alarm system	garage	FALSE
204	camera	garage	FALSE
205	camera	living room	TRUE

a) Create the `security_device` table. **5**

b) Update the `security_device` table to include a new column named `battery_level` to track the battery percentage of each device. The `battery_level` should be stored as an integer. **2**

c) Set the `status` of the device with `device_id` 203 to 'Active'. **3**

Total 10 marks

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Marks

Question 7

The smart home security system contains the following tables.

user

user_id	first_name	last_name	contact_number
101	John	Smith	1234567
102	Aidan	Johnson	2345678
103	Sarah	Lee	3456789

security_device

device_id	device_type	location	status (TRUE=Active, FALSE=Inactive)	battery_level (%)
201	camera	entrance	TRUE	75
202	motion sensor	living room	TRUE	30
203	alarm system	garage	FALSE	50
204	camera	garage	FALSE	45
205	camera	living room	TRUE	60

access_log

log_id	access_time	user_id	device_id(FK)
301	2025-12-23 09:00:00	101	201
302	2025-04-02 12:00:00	102	202
303	2025-10-14 08:00:00	101	203
304	2025-09-20 19:00:00	102	203
305	2025-09-16 11:00:00	102	202

a) Write an SQL command to list all access logs for devices of type “camera”. **3**

b) Write an SQL command to display each user’s name, contact number and the most recent time they accessed a security device. **4**

c) Write an SQL command to list all security devices where the battery level is below 50%. **3**

Total 10 marks

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Marks

Question 8

A smart home security installation company manages customer information, installation schedules and maintenance requests. Currently, the company relies on spreadsheets and paper-based logs to keep track of customer records, installed security devices and service appointments. **10**

As the company expands, it faces numerous challenges related to its record system. The company is now considering implementing a database system to improve efficiency.

Identify and explain **FIVE (5)** benefits of implementing a database system over their current spreadsheet and paper-based record-keeping system.

Total 10 marks

Marks

Question 9

a) A smart home security monitoring company provides 24/7 surveillance and remote access to security cameras, alarms and motion sensors for homeowners and businesses. The company relies on a central database to store real-time security footage, device status, access logs and customer account details. **8**

However, during a recent database failure, it was discovered that no recent backups existed and the recovery process was inadequate. This caused major disruptions in service, impacting both customers and the company's operations.

Discuss **FOUR (4)** key areas where the lack of proper database backup and recovery processes can negatively impact the smart home security monitoring company. For each area, include a specific example of how the failure could affect customers or business operations.

b) A smart home installation company provides customised smart security systems for private residences. Their systems include smart locks, security cameras, motion sensors and alarm systems, all controlled through a central system. **2**

Since they deal with sensitive security data, the company wants full control over their database to ensure data privacy, security and custom system integration.

Identify a suitable cloud database model for the company. Provide **ONE (1**) reason to support your answer.

Total 10 marks

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Marks

Question 10

A global e-commerce company is developing a backend system to manage its supply chain and logistics operations. The system must handle:

- Real-time order tracking, ensuring customers can see the status of their deliveries.
- Customer purchase history and preferences, allowing for personalised recommendations.

To meet these requirements, the company is considering adopting **NoSQL** databases for efficient scalability and performance.

a) Identify **TWO (2)** NoSQL data models that could be used to store different types of data for the platform. Provide **ONE (1)** example of data that each model would handle best. **4**

b) Identify the **THREE (3)** elements of the 3V model. **6**

Explain how each element applies to the data challenges of the e-commerce logistics platform.

Total 10 marks

End of Paper

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